Systems Analysis and Control

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Lecture 9: Dynamics of Response: Complex Poles

Overview

In this Lecture, you will learn:

Characteristics of the Response

Complex Poles

- Rise Time
- Settling Time
- Percent Overshoot

Performance Specifications

- Geometric Pole Restrictions

Relationship between $\omega_d, \sigma, \omega_n, \zeta$ for Complex Poles

Frequency and Damping terms

3 Different Forms: Each with $y_{ss} = 1$

$$\hat{G}(s) = \frac{b}{s^2 + as + b}$$

$$\hat{G}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\hat{G}(s) = \frac{\sigma^2 + \omega_d^2}{s^2 - 2\sigma s + (\sigma^2 + \omega_d^2)} = \frac{\sigma^2 + \omega_d^2}{(s - \sigma + \imath\omega_d)(s - \sigma - \imath\omega_d)} = \frac{\sigma^2 + \omega_d^2}{(s - \sigma)^2 + \omega_d^2}$$

Two Complex Poles at $s = \sigma \pm \imath\omega_d$.

- **Damped Frequency:** ω_d
  - $\omega_d = \sqrt{b - \frac{a^2}{4}} = \omega_n\sqrt{1 - \zeta^2}$
- **Decay Rate:** σ
  - $\sigma = -\frac{a}{2} = -\zeta\omega_n$
- **Natural Frequency:** ω_n
  - $\omega_n = \sqrt{b} = \sqrt{\sigma^2 + \omega_d^2}$
- **Damping Ratio:** ζ
  - $\zeta = \frac{a}{2\omega_n} = \frac{|\sigma|}{\omega_n} = \frac{|\sigma|}{\sqrt{\sigma^2 + \omega_d^2}}$

Step Response for Complex Poles

$$\hat{y}(s) = \frac{\omega_d^2 + \sigma^2}{s^2 + 2\sigma s + \omega_d^2 + \sigma^2}\frac{1}{s} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}\frac{1}{s} = \frac{k_1 s + k_2}{s^2 + 2\zeta\omega_n s + \omega_n^2} + \frac{r_2}{s}$$

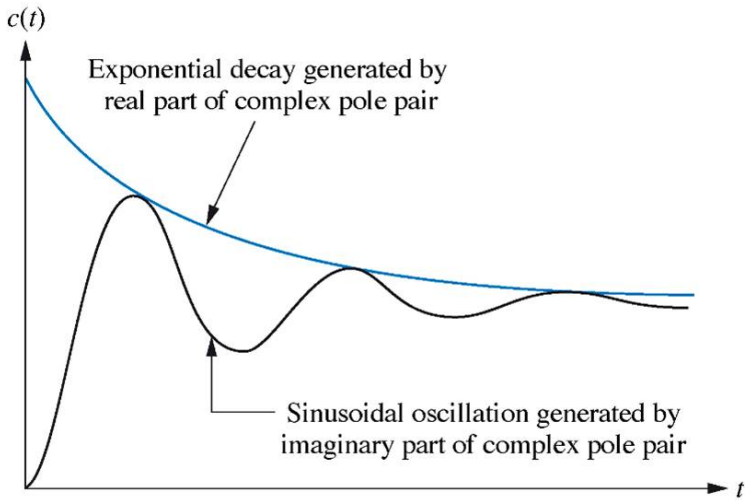
The poles of $\hat{y}$ are at $s = \sigma \pm \omega_d \imath$. Using PFE, the solution is:

$$\begin{aligned} y(t) &= 1 - e^{\sigma t}\left(\cos(\omega_d t) + \frac{\sigma}{\omega_d}\sin(\omega_d t)\right) \\ &= 1 - e^{\sigma t}\frac{\omega_n}{\omega_d}\sin\left(\omega_d t + \phi\right) \end{aligned}$$

Where $\sigma = \zeta\omega_n$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $\phi = \tan^{-1}\left(\frac{\omega_d}{\zeta\omega_n}\right)$.

The result is oscillation with an Exponential Envelope.

- Envelope decays at rate σ
- Speed of oscillation is ω_d, the **Damped Frequency**



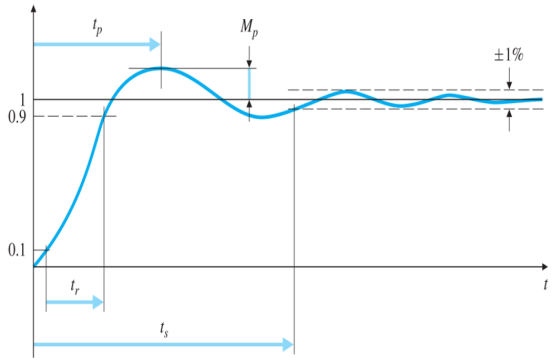
Rise Time (T_r) for Complex Poles

An Approximate Formula for Rise Time

Recall: T_r is the time to go from .1 to .9 *of the final value*

Problem: T_r increases with **Damping**.

- Damping *slows* the response.
- Variation with ζ, ω_n is *Nonlinear*!
- No accurate formula available.



Damping ratio	Normalized rise time
0.1	1.104
0.2	1.203
0.3	1.321
0.4	1.463
0.5	1.638
0.6	1.854
0.7	2.126
0.8	2.467
0.9	2.883

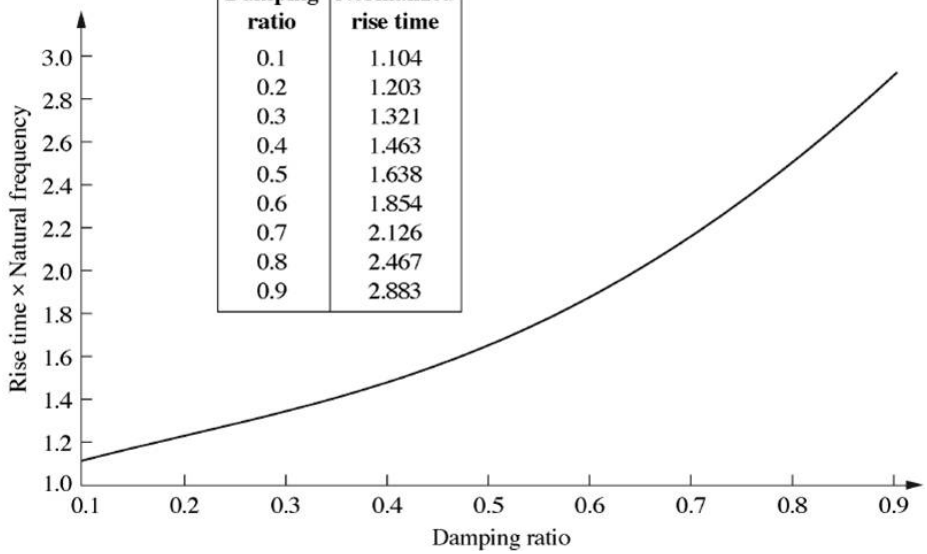


Figure: Plot of $T_r \cdot \omega_n$ for different values of Damping, ζ

In this class, when asked for rise-time for a complex pole, use the formula:

$$T_r \cong \frac{1.8}{\omega_n}$$

We randomly chose $\zeta = .55$

Note: This is a total hack, but one which everybody uses....

Settling Time (T_s) for Complex Poles

Recall: T_s is time it takes to reach and stay within .99 of final value.
Recall the step response

$$y(t) = 1 - e^{\sigma t}\left(\cos(\omega_d t) + \frac{\sigma}{\omega_d}\sin(\omega_d t)\right)$$

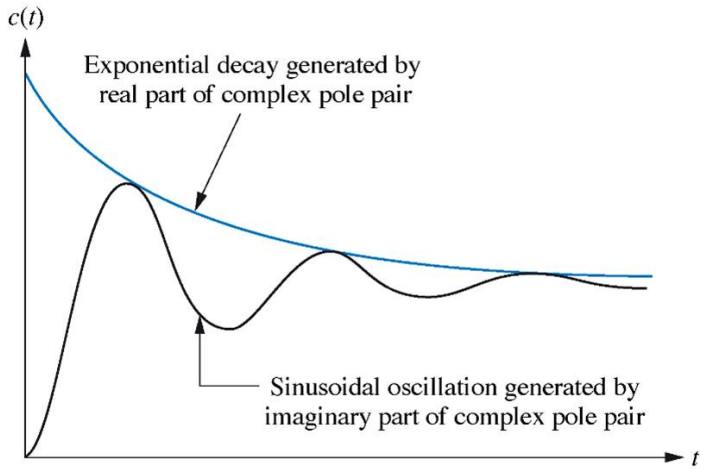
Oscillations are confined within an *Exponential Envelope*

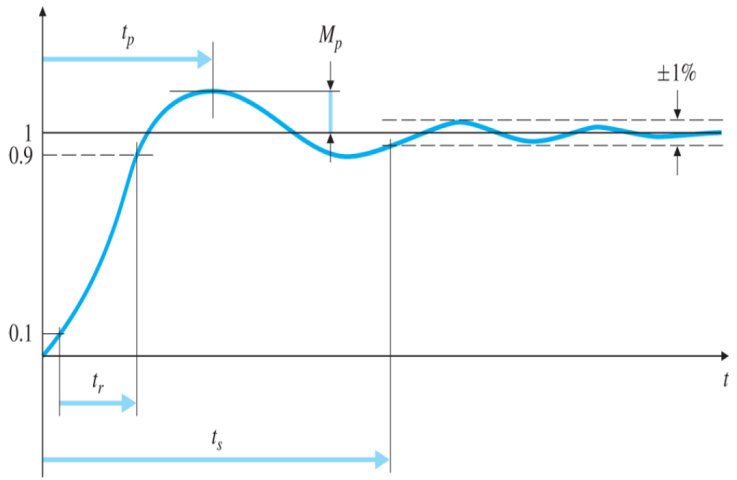
- The exponential envelop decays at rate σ.
- **Settling Time** for a complex pole is given by contraction of the envelope

$$\|1 - y(t)\| \leq e^{\sigma t} \leq .01$$

For a complex pole (same as for real pole), we have the formula:

$$T_s \cong \frac{4.6}{-\sigma}$$

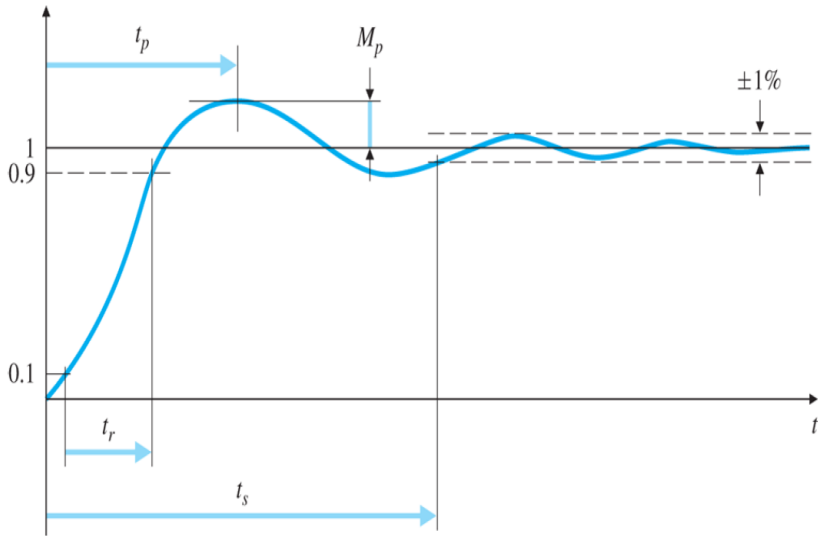




Peak Time for Complex Poles

The time of MAXIMUM deflection.

- Only Complex poles have peaks.



Definition 1.

The **Peak Time**, T_p is time at which the step response obtains its maximum value.

To calculate T_p, we must find when

$$y(t) = 1 - e^{\sigma t}\left(\cos(\omega_d t) + \frac{\sigma}{\omega_d}\sin(\omega_d t)\right) = 1 - e^{\sigma t}\frac{\omega_n}{\omega_d}\sin\left(\omega_d t + \phi\right)$$

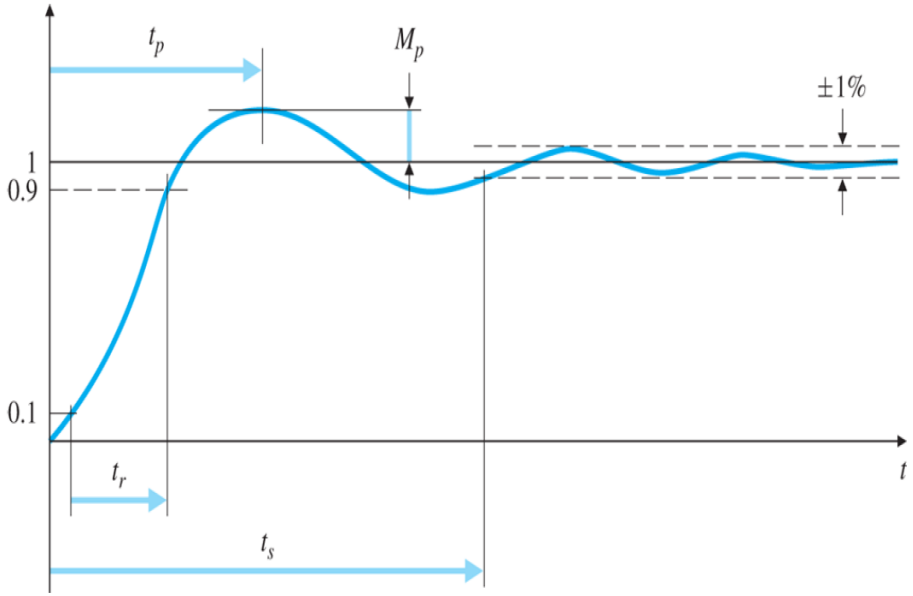
Achieves its maximum.

Formula for Peak Time (T_p) for Complex Poles

To find the extrema, we set $\dot{y}(t) = 0$, where it can be shown that

$$\dot{y}(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{\sigma t} \sin \omega_d t$$
$$= \frac{\omega_n^2}{\omega_d} e^{\sigma t} \sin \omega_d t$$

So $\dot{y}(t) = 0$ when $t = \frac{n\pi}{\omega_d}$.

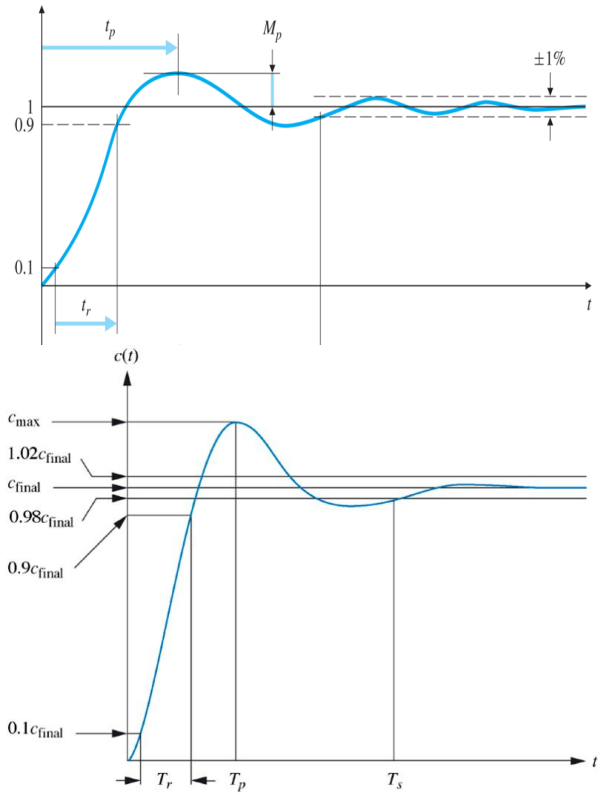


Because of the exponential envelope, the first peak will always be largest.

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

Percent Overshoot (M_p) for Complex Poles

Unique to complex poles is the concept of overshoot:



Definition 2.

The **Percent Overshoot**, M_p is the peak value of the signal, as a percentage of steady-state ($\frac{c_{\max}-c_{final}}{c_{final}} \cdot 100\%$).

To find M_p, we need the max of $y(t)$ (We need $(c_{\max})$).

- $c_{\max} = y(T_p)$ where $T_p = \frac{\pi}{\omega_d}$.

Systems with high overshoot may move violently before settling.

- Can cause crashes, un-modeled dynamics, etc.

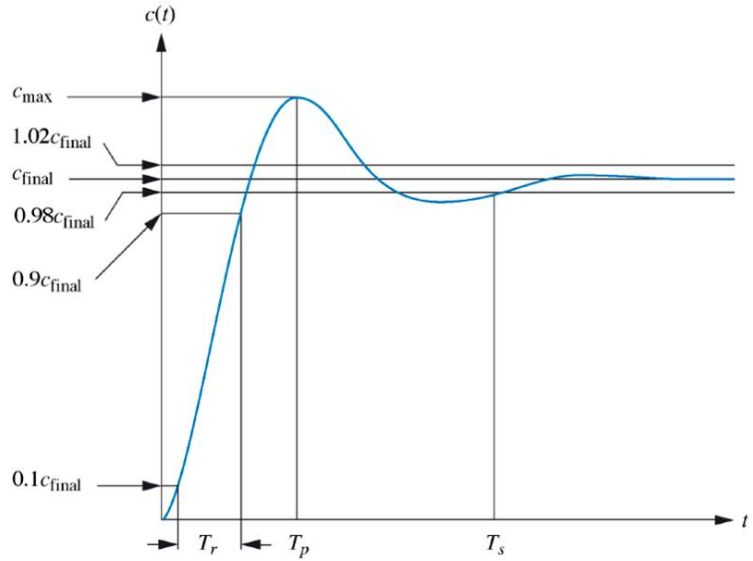
Formula for Percent Overshoot (M_p) for Complex Poles

To calculate M_p, we need the maximum value of $y(t)$.

- Occurs at time $T_p = \frac{\pi}{\omega_d}$.

Since $c_{final} = 1$ ($y_{ss} = 1$):

$$\begin{aligned} M_p &= y(T_p) - 1 \\ &= -e^{\sigma T_p}\left(\cos(\omega_d T_p) + \frac{\sigma}{\omega_d}\sin(\omega_d T_p)\right) \\ &= -e^{\sigma T_p}\left(\cos(\pi) + \frac{\sigma}{\omega_d}\sin(\pi)\right) \\ &= e^{\sigma T_p} = e^{\frac{\pi\sigma}{\omega_d}} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \end{aligned}$$



$$M_p = e^{\frac{\pi\sigma}{\omega_d}} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

M_p depends only on ζ!

Complex Poles

Lab Example

Estimate:

- Rise/Peak Time
- Percent Overshoot

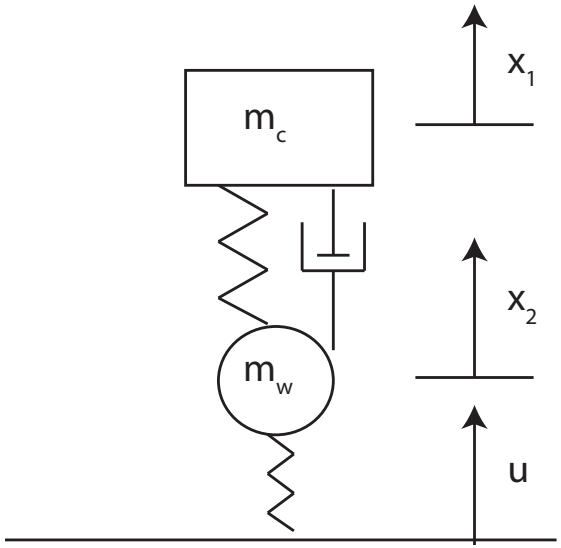
Computing T_r, T_s, T_p, M_p for Complex Poles (Open Loop)

Numerical Example: Find $\omega_d, \sigma, \omega_n, \zeta$ for both sets of poles

Lets look at the suspension problem

Open Loop:

$$\hat{G}(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$



The poles are:

- $p_{1,2} = -.9567 \pm 1.2272\imath$
- $p_{3,4} = -.0433 \pm .6412\imath$

Because there are two sets of poles, we should consider both.

$$\sigma_1 = -.9567 \qquad \omega_{d,1} = 1.2272$$
$$\sigma_2 = -.0433 \qquad \omega_{d,2} = .6412$$

$$\omega_{n,1} = \sqrt{\sigma_1^2 + \omega_{d,1}^2} = 1.5561 \qquad \zeta_1 = \frac{|\sigma|}{\omega_{n,1}} = .6148$$

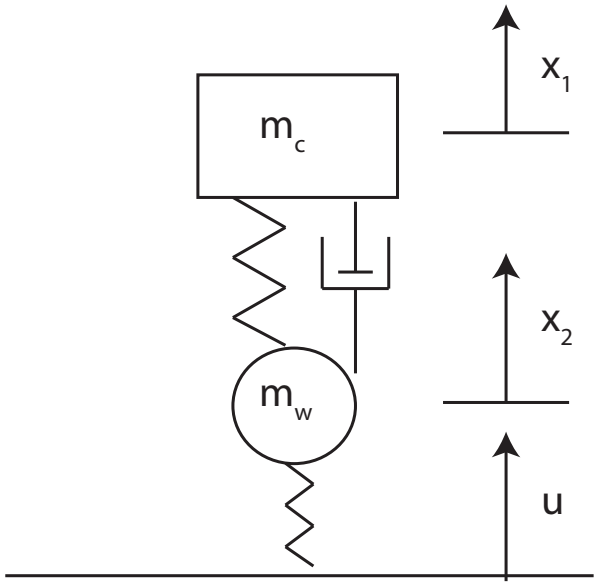
$$\omega_{n,2} = .6427 \qquad \zeta_2 = .0674$$

Computing T_r, T_s, T_p, M_p for Complex Poles (Closed Loop)

Numerical Example: Find $\omega_d, \sigma, \omega_n, \zeta$ for both sets of poles

Closed Loop (Upper Feedback): Let $k = 1$

$$\hat{G}(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + (3+k)s^2 + (1+k)s + (1+k)}$$



The poles are:

- $p_{1,2} = -.8624 \pm 1.4391\imath$
- $p_{3,4} = -.1376 \pm .8316\imath$

Consider both sets of poles.

$$\sigma_1 = -.8624 \qquad \omega_{d,1} = 1.4391$$
$$\sigma_2 = -.1376 \qquad \omega_{d,2} = .8316$$

The natural frequency and damping ratios are

$$\omega_{n,1} = 1.6777 \qquad \zeta_1 = .5140$$

$$\omega_{n,2} = .8429 \qquad \zeta_2 = .1632$$

Computing T_r, T_s, T_p, M_p for a Complex Pole

Numerical Example: Percent Overshoot

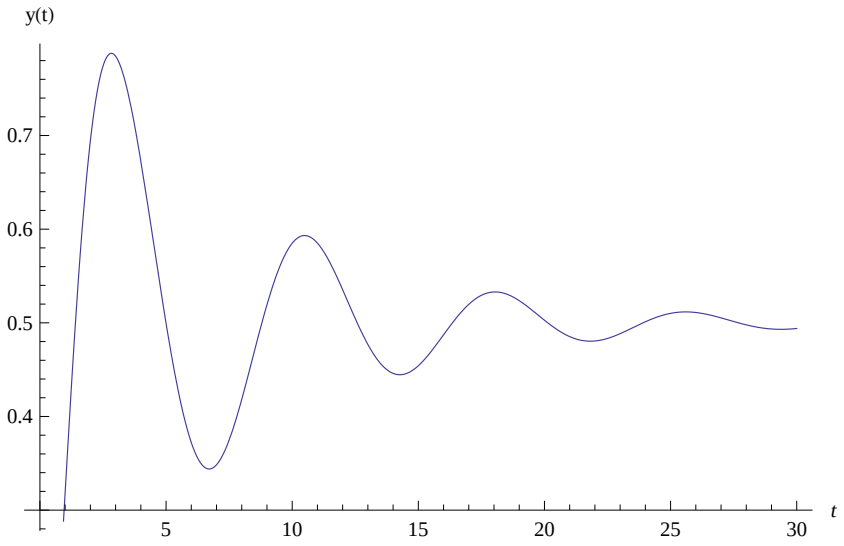


Figure: Closed Loop

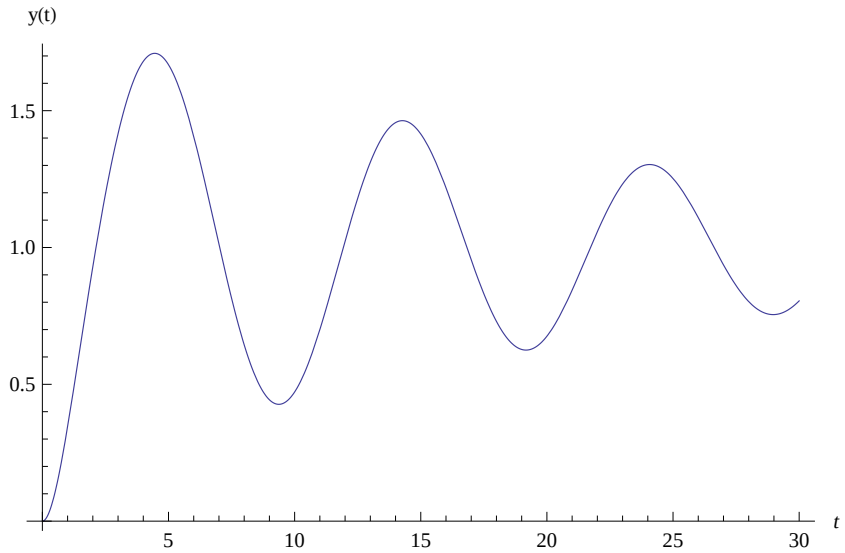


Figure: Open Loop

Overshoot: Open Loop (easiest to use σ and ω directly)

$$M_{p,1} = e^{\frac{\pi\sigma_1}{\omega_1}} = .0864 \qquad M_{p,2} = .8088$$

Overshoot: Closed Loop (Upper Feedback)

$$M_{p,1} = .152 \qquad M_{p,2} = .5946$$

A substantial improvement in performance.

Computing T_r, T_s, T_p, M_p for a Complex Pole

Numerical Example: Settling Time

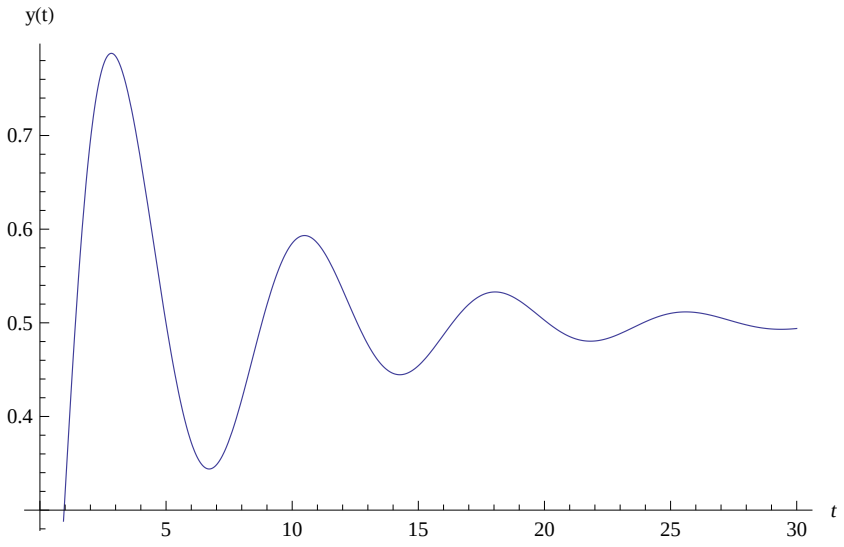


Figure: Closed Loop

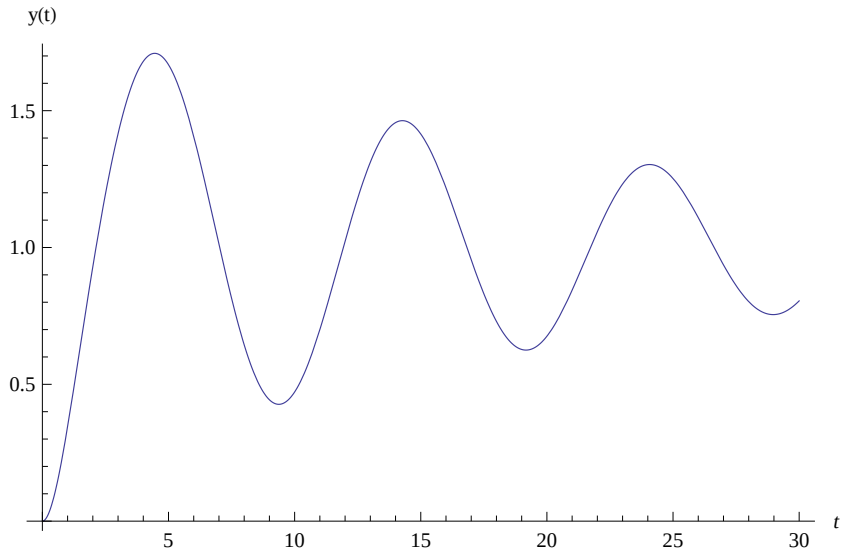


Figure: Open Loop

Settling Time: Open Loop

$$T_{s,1} = \frac{4.6}{-\sigma} = 4.81 \qquad T_{s,2} = 106.23$$

Settling Time: Closed Loop (Upper Feedback)

$$T_{s,1} = 5.33 \qquad T_{s,2} = 33.43$$

Computing T_r, T_s, T_p, M_p for a Complex Pole

Numerical Example: Peak Time

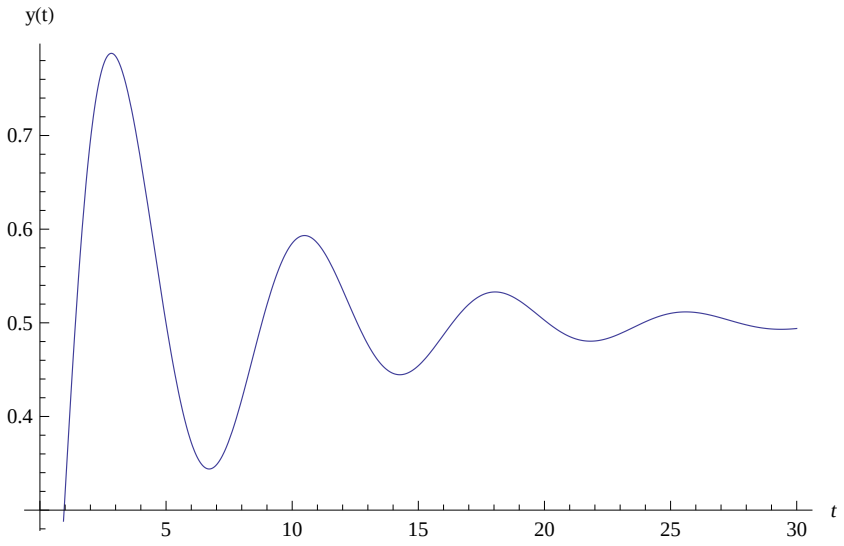


Figure: Closed Loop

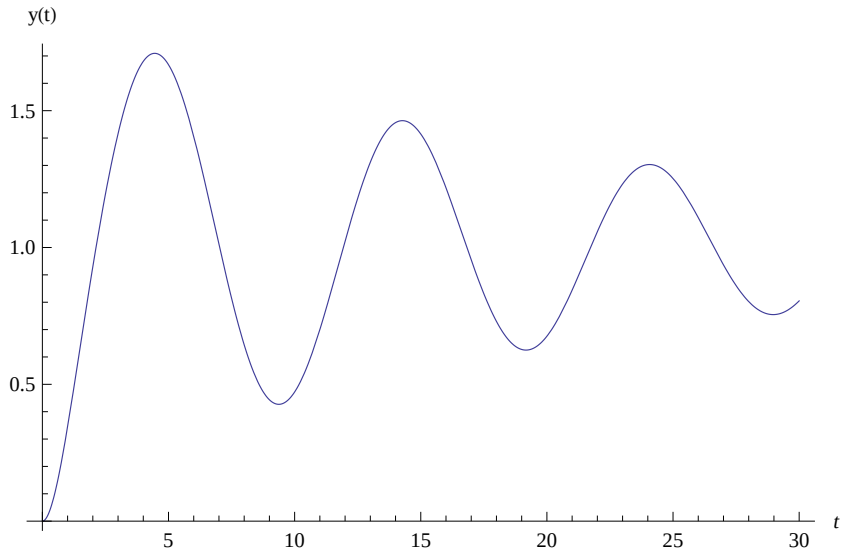


Figure: Open Loop

Peak Time: Open Loop

$$T_{p,1} = \frac{\pi}{\omega_d} = 2.56 \qquad T_{p,2} = 4.90$$

Peak Time: Closed Loop

$$T_{p,1} = 2.18 \qquad T_{p,2} = 3.78$$

Performance Specifications

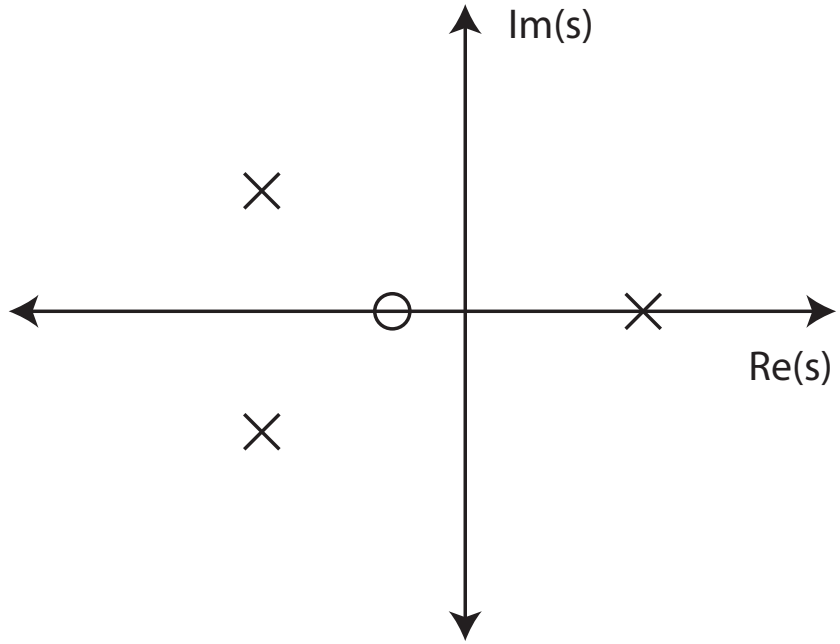
Interpretation as Geometric Constraints on Pole Locations

Dynamic response is determined by *pole locations*.

- Except Steady-State Error

Usually, dynamic response **improves** with feedback.

- Recall the numerical examples.
- Pole locations change under feedback.
- The choice of $k = 1$ was just a guess.



Consider: The goal of a controller is to change the location of the poles.

- But where do we want them?

Performance Specifications create Geometric Constraints in the Complex Plane.

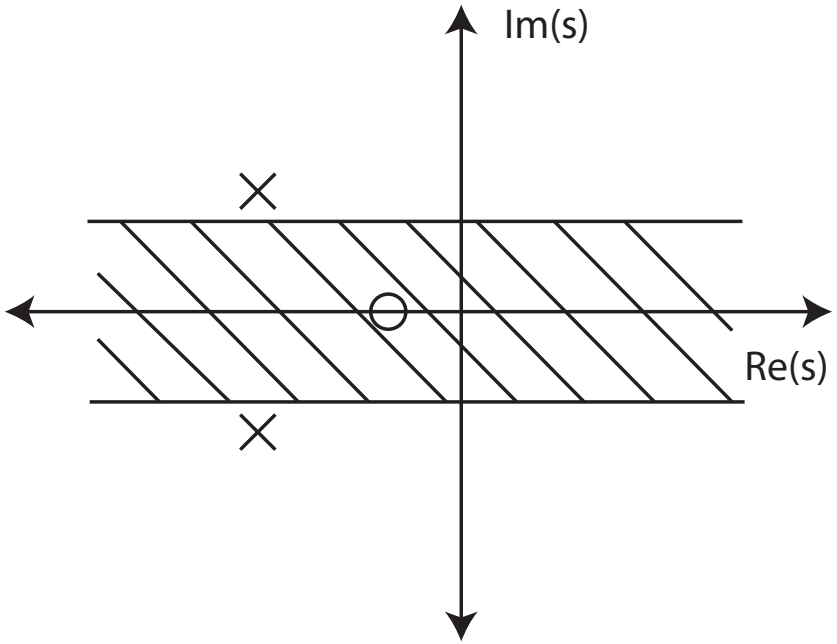
Peak Time Constraints on Pole Locations

Suppose we have performance specs for

- T_p, T_r, M_p etc.

We can translate this to regions of the complex plane.

Maximum Peak Time: $T_{p,desired}$.



We usually require $T_p < T_{p,desired}$.

$$\frac{\pi}{\omega_d} = T_p < T_{p,desired}$$

Which means

$$\omega_d > \frac{\pi}{T_{p,desired}}$$

The geometric interpretation is that the imaginary part be sufficiently large.

Settling Time Constraints on Pole Locations

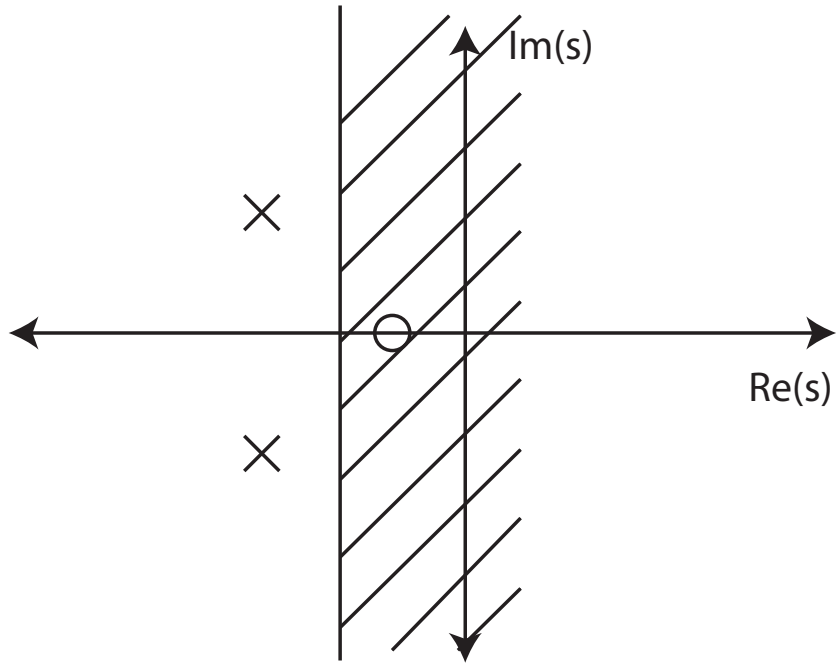
Maximum Settling Time: $T_{s,desired}$.

We want quick convergence.

- So we require $T_s < T_{s,desired}$.

Hence,

$$\frac{4.6}{-\sigma} = T_s < T_{s,desired}$$



Which translates to

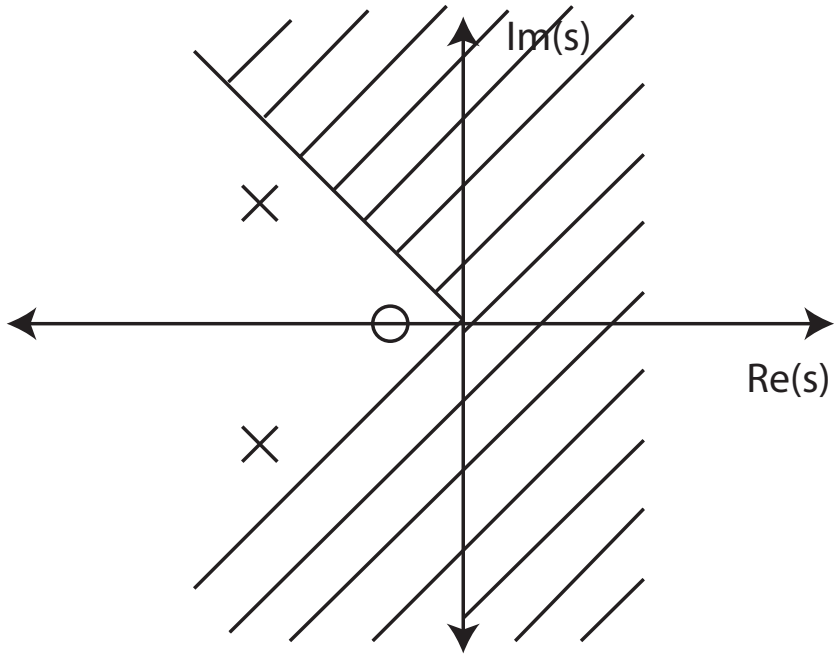
$$\sigma < -\frac{4.6}{T_{s,desired}}$$

The geometric interpretation is that the real part be sufficiently negative.

Percent Overshoot (M_p) Constraints on Pole Locations

Interpretation using ω_d and σ

Maximum Overshot: $M_{p,desired}$.



We don't like hitting things,

- So we need $M_p < M_{p,desired}$.

2 roots $(\sigma \pm \omega_d \imath)$ give two constraints:

$$e^{\frac{\pi\sigma}{\pm\omega_d}} = M_p < M_{p,desired}$$

$$\frac{\pi\sigma}{\pm\omega_d} < \ln(M_{p,desired})$$

or since $\ln(M_{p,desired}) < 0$,

$$\omega_d < \frac{\pi}{\ln(M_{p,desired})}\sigma, \qquad \omega_d > -\frac{\pi}{\ln(M_{p,desired})}\sigma$$

A sector constraint on σ and ω?

Percent Overshoot (M_p) Constraints on Pole Locations

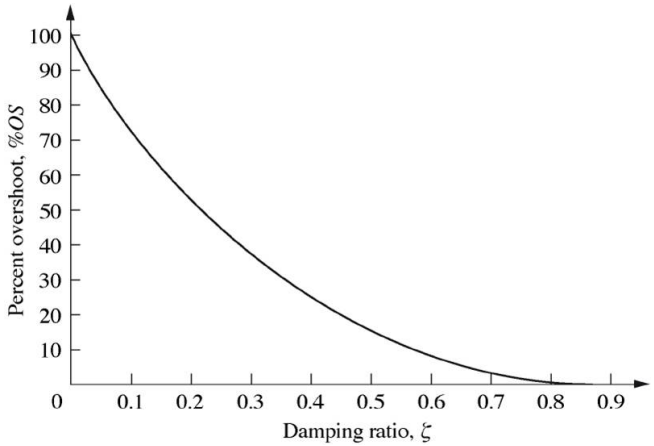
Interpretation using Damping Ratio ζ

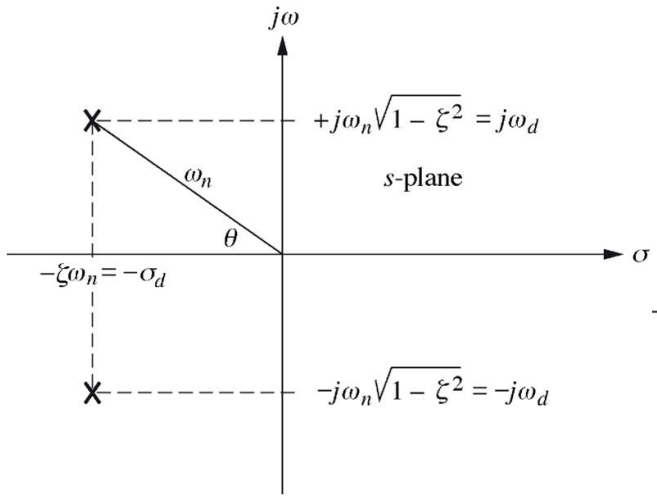
$$\hat{G}(s) = \frac{b}{s^2 + as + b} \qquad \hat{G}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

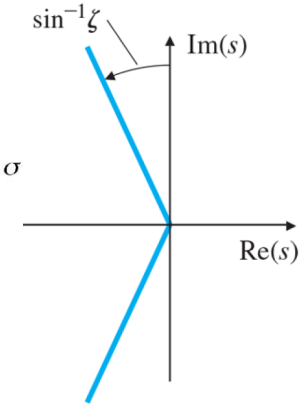
M_p is determined by damping ratio ALONE:

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}, \qquad \rightleftarrows \qquad \zeta = \sqrt{\frac{\ln(M_p)^2}{\pi^2 + \ln(M_p)^2}}$$

Interpret this as an ANGLE in the complex plane:





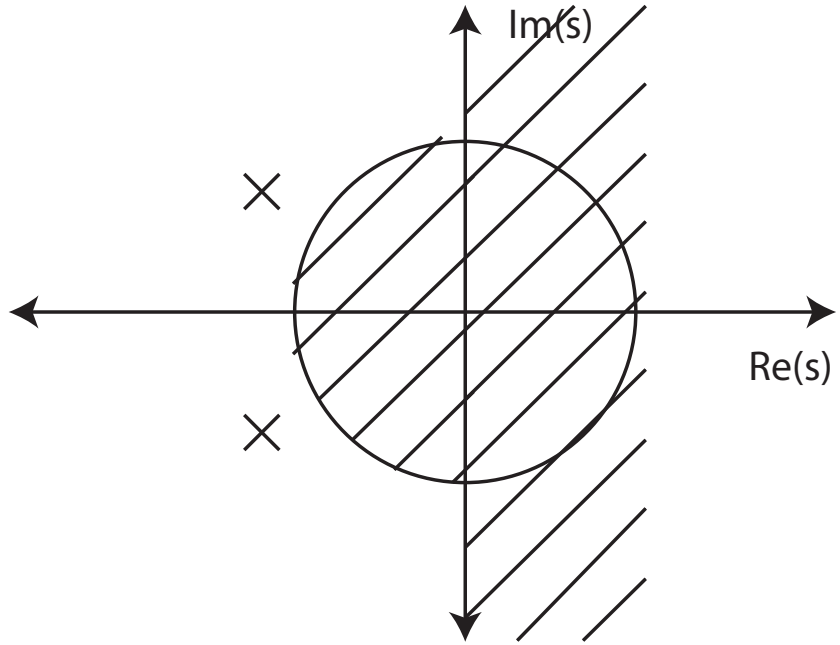


where recall: $\zeta = \frac{a}{2\omega_n} = \frac{|\sigma|}{\omega_n} = \frac{|\sigma|}{\sqrt{\sigma^2+\omega_d^2}}$

Rise Time T_r Constraints on Pole Locations

The expression for rise time is complicated. We use $\zeta = .5$, to get

$$T_r \cong \frac{1.8}{\omega_n}$$



Maximum Rise Time: $T_{r,desired}$.

- We want quick response.
- We require $T_r < T_{r,desired}$.

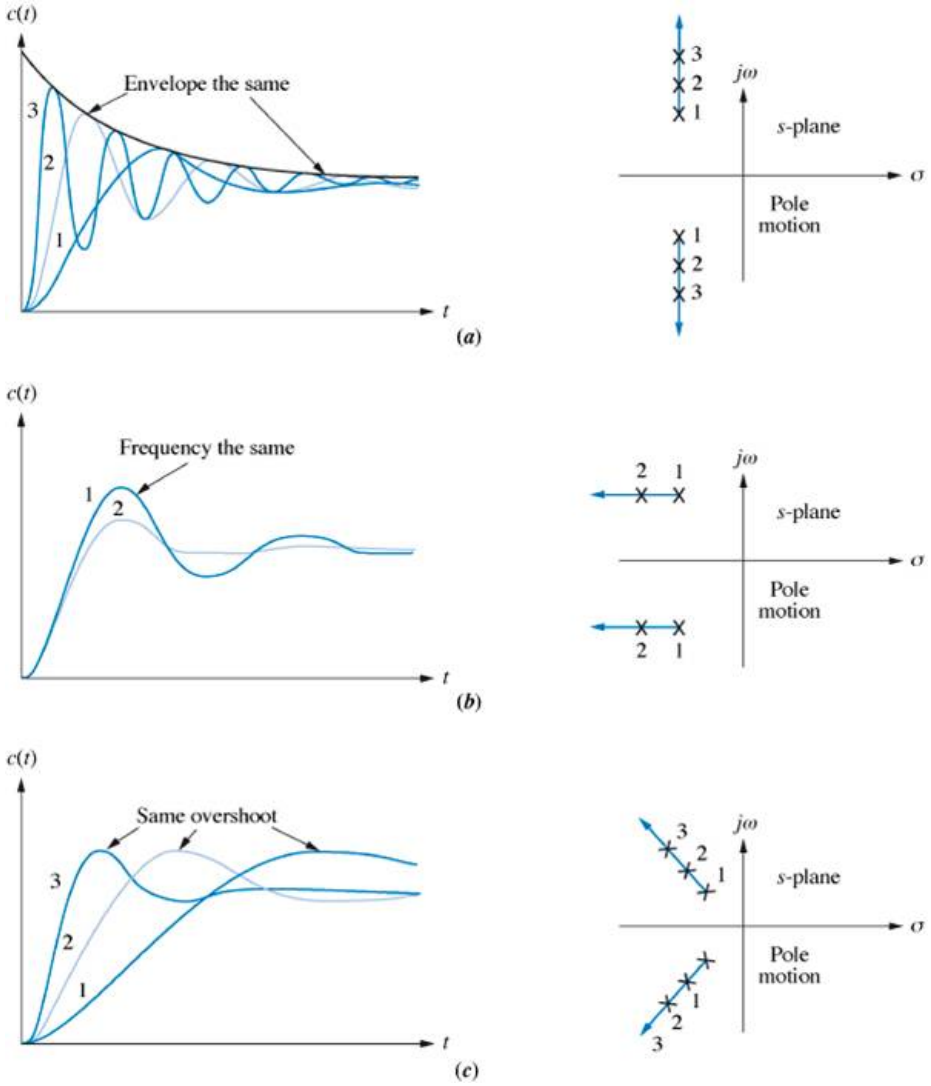
$$\frac{1.8}{\omega_n} = T_r < T_{r,desired}$$

Thus we require

$$\omega_n > \frac{1.8}{T_{r,desired}}$$

Recall that $\omega_n = \sqrt{\sigma^2 + \omega_d^2}$, so the geometric interpretation is a circle: $\|s\| > \frac{1.8}{T_{r,desired}}$.

Effect of Changes in ω_d, σ, and ζ



Constraints from Multiple Performance Specifications

Combined Restrictions on the Complex Plane

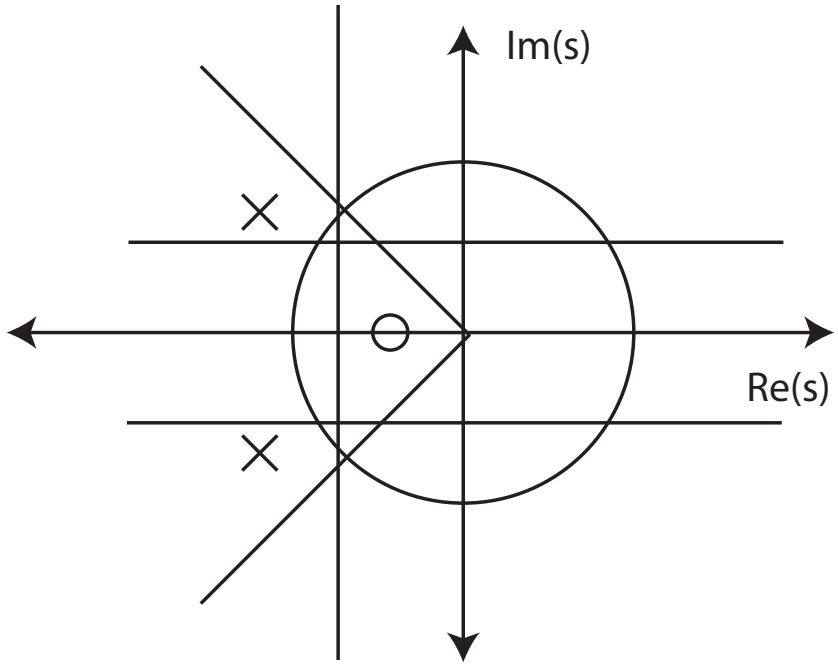
Mostly, we have several constraints

$$\omega_d > \frac{\pi}{T_{p,desired}}$$

$$\sigma < -\frac{4.6}{T_{s,desired}}$$

$$\omega_d < \frac{\pi}{\ln(M_{p,desired})}\sigma$$

$$\omega_n > \frac{1.8}{T_{r,desired}}$$



NOTE: There will be MANY pole locations which meet the specifications

- Any pole locations which are not prohibited are OK.

Selecting Poles to Fit Multiple Constraints

A Numerical Illustration: Second Order System

High Performance Aircraft: Select Pole Locations $p = \sigma \pm \imath\omega_d$ such that

- **Overshoot:** Reduce overshoot to less than 5%.

$$M_{p,desired} = .05$$

$$\omega_d < \frac{\pi}{\ln(M_{p,desired})}\sigma = -1.05|\sigma|$$

- A difficult requirement to meet?
- **Rise Time:** Quick response is critical. Limit Rise Time to 1s or less

$$T_{r,desired} = 1$$

$$\omega_n > \frac{1.8}{T_{r,desired}} = 1.8$$

- **Settling Time:** Limit settling time to $T_{s,desired} = 4.6s$.

$$T_{s,desired} = 4.6s$$

$$\sigma < -\frac{4.6}{T_{s,desired}} = -1$$

Poles Constraints From Performance Specifications

Two or Three Specifications?

NOTE: For any 2 specifications, we can find a pole location which meets the specifications, EXACTLY.

EXAMPLE:

- For $T_s = 4.6$, we have $\sigma = -1$.
- Then for $M_p = .05$, we have $\omega_d = 1.05|\sigma| = 1.05$

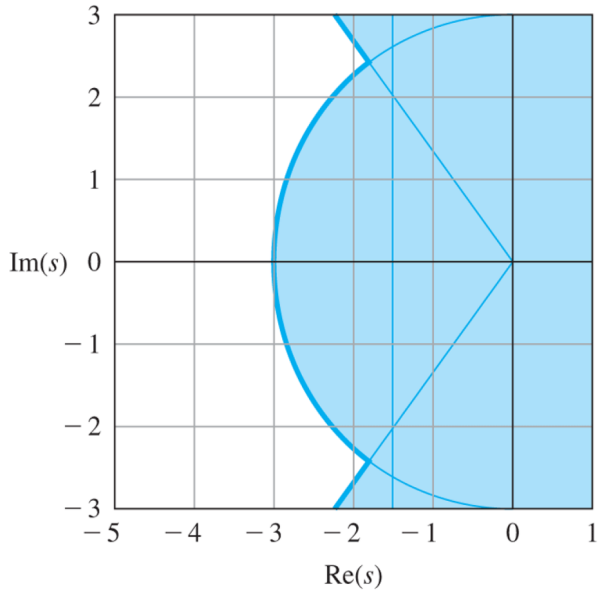
HOWEVER:

- Now, $\omega_n = \sqrt{\sigma^2 + \omega_d^2} = 1.45$
- So $T_r = \frac{1.8}{\omega_n} = 1.24$! Too Slow.

ISSUE: For 3 specifications, we may need to IMPROVE one of the other specs.

SOLUTION: Increase T_s – lets try $T_s = 3$.

- Then $\sigma = -1.533$
- Then from M_p specification, choose $\omega_d = 1.05 \cdot 1.533 = 1.61$
- Then $\omega_n = \sqrt{\sigma^2 + \omega_d^2} = 2.22$
- So $T_r = \frac{1.8}{\omega_n} = .8096$! Fast Enough!



Pole Locations

Missile Video

Estimate Performance Specs:

Summary

What have we learned today?

In this Lecture, you will learn:

Characteristics of the Response

Complex Poles

- Rise Time
- Settling Time
- Percent Overshoot

Performance Specifications

- Geometric Pole Restrictions

Next Lecture: Designing Controllers